HTML

HTML stands for Hyper Text Markup Language. HTML is the standard markup language for creating web pages. HTML describe the structure of web page. HTML consists of a series of elements. HTML elements tell the browser how to display the content.

→ Basic Structure of HTML 5

```
<!DOCTYPE html>
<head>
<title> Page Title </title>
</head>
<body>

    <h1> My First Heading</h1>
    <P> My First Paragraph. </P>
</body>
</html>
```

Structure Explaination.

The <!DOCTYPE html> declaration defines that this document is an HTML 5 document.

<html> element is the root element of an HTML Page.

<head> element contains meta information about the HTML page.

<title> element specifies a title for the HTML page.

The <body> element defines the document's body, and is a container for all visible contents, such as headings, paragraph, images etc.

<P> element defines a paragraph.

<h1> element defines a large headings.

→ Heading Tag:-

A heading tag is used to define headings or titles on a web page. They help organize content and make it easy to read. Headings improve website structure.

Heading Tags:-

HTML Provides Six heading tags.

<h1> → Main heading of the page (most important & and largest in size.

<h2> → Subheading

<h3> → Smaller heading and subheading

<h4> → less important heading

<h5> → Minor heading with low importance.

<h6> → Smallest and least important heading

→ Strong Tag:-

The tag is used to show important text in HTML. It makes the text bold by default. It gives semantic meaning (importance) to the content. Browser and screen readers treat it as important information. It helps in SEO and accessibility.

Example:-

<P> This is a <Strong> very important </Strong> message. </P>

Emphasis Tag:-

The (emphasis) tag in HTML is used to emphasize or stress a word in a sentence. The text inside this tag is usually shown in italic style by the browser. It gives semantic meaning that the text is important in the context. Screen readers read it with emphasis.

Unordered List:-

The unordered list <ul> tag is used to create an unordered list with no specific order. Items are shown with bullet points (•). Each item is written inside <li> tag. It is used when order does not matter.

Example:-

```
<ul>
<li> Apple </li>
<li> Banana </li>
<li> Mango </li>
</ul>
```

Output:-

- Apple
- Banana
- Mango

→ **Orderlist (<ol>):-**

The ordered list in HTML is used to display a list of items in a specific sequence or order. Items are shown with numbers (1, 2, 3.....). Its is used when steps or order is important for example instructions, steps, ranking. It is created using the <ol> tag, and each list item is written inside the <li> tag.

→ **Example:-**

```
<ol>
<li> Wake up <\li>
<li> Study <\li>
<li> Sleep <\li>
<\ol>
```

Output:-

1. Wake up
2. Study
3. Sleep

Anchor Tag (<a>):-

The <a> anchor tag in HTML is used to create hyperlinks that connect one webpage to another page, file, email, address, or location within the same pages. It uses the href attribute to specify the link's destination

Example:-

```
<a
href = "https://www.google.com"> visit
this website </a>
```

Imag Tag:-

<img> tag is used to display images in a web page. It is an empty tag (no closing tag). In the <img> tag src attribute defines the image path. alt attribute provides image description

Example:-

```
<img src="image.jpg" alt="Sample
Image">
```

Video Tag:-

The HTML <video> tag is used to embed video content on a web page. It allows you to play a video files directly in the browser without external plugins. You can control playback using attribute like controls, autoplay, loop and muted.

Example:-

```
<video width="320" height="240"
controls>
<source src="movie.mp4" type="video/mp4">
your Browser does not support this video
tag. </video>
```

→ ## Form Tag:-

The <form> tag in HTML is used to create a form for collecting user input (such as test passwords, emails etc). It sends the entered data to a server for processing.

It commonly uses attribute like action (where data is sent) and method (how data is sent).

Example:-

```
<form action="submit.php" method="post">
<input type="text" name="username">
</form>
```

Span Tag:-

 tag is an inline container used to group text or elements for styling or scripting purposes. It's often used with CSS or JavaScript to apply styles or manipulate specific parts of content.

Table Tag:-

The <table> tag in HTML is used to create a table structure to display data in rows and columns.

- <tr> (table row) - Defines a row in the table.
- <th> (table heading) it defines a headings of table.
- <td> (table data) this tag is used to define a table cell in a table row (<tr>)

Structure of Table:-

```
<table
<tr>
 <th> Name </th>
 <th> Contact </th>
 <th> Roll no </th>
</tr>
 <td> Ali </td>
 <td> Email </td>
 <td> 123 </td>
 </table>
```

Semantic Elements:

Semantic elements in HTML are elements that clearly describe their meaning or purpose both to the browser and developers. They improve readability, accessibility and SEO by giving context to the content.

Example:-

<header> , <footer> , <article>
<section>

checkbox:-

checkbox (<input type="checkbox">) allows the user to select one or more options from a set.

Placeholder:-

Placeholder (placeholder attribute) shows hint text inside on input field before the user types.

<input type="text"
Placeholder="Enter your name">

Select:-

<select> creates a dropdown list of options for the user to choose form.

Radio Button:-

<input type="radio"> allows the user to select only one option from a group.

Example:-

<input type="radio" name="gender"
value="male">